Metrics, evaluation and launch thresholds

Extracted from Sections 3.7, 5.3, 5.4, 5.6 and 5.7 of the final report.

Ground truth, built three ways

1. Implicit labels — every reviewer decision (signed off, sent back, escalated) is a label attached to a report, produced by work that would happen anyway.
2. Explicit labels — a stratified sample of 30 reports per week, labelled independently by an experienced reviewer against the same written severity rubric given to the model, before seeing the model output. This is the evaluation set of record.
3. Correction labels — every override with its reason, where systematic mistakes become visible as patterns rather than anecdotes.

The rubric is versioned in the same repository as the prompt. A disagreement between two human labellers is treated as a rubric defect to be fixed, not noise to be averaged away.

Baselines run in parallel

Baseline	What it represents	Why it matters
Chronological queue	Today's behaviour — reports read in arrival order	If AI ordering does not beat arrival order, the feature has no reason to exist
Rules-only ordering	Numeric thresholds and keyword flags, buildable in about a week	The honest bar. AI must add value beyond what simple rules catch, specifically on issues expressed only in prose

Evaluation framework

Type	Metric	Why it matters
Technical	Severity recall on serious (P1) issues	Missing a serious issue is the failure that hurts; recall is prioritised over precision
Technical	Severity classification precision overall	Too many false alarms and reviewers stop trusting the ordering
Technical	Issue extraction F1 against the weekly labelled sample	Whether the layer sees what a careful human reader sees
Technical	Groundedness — share of summary statements traceable to a source sentence	The direct measure of hallucination in this product
Technical	Schema validation failure rate and median latency	The queue must be ready when the reviewer opens it
Business	Reviewer minutes per campus report	The efficiency claim, measured rather than asserted
Business	Time from issue written to issue	The outcome the institution actually

	escalated	cares about
Business	Send-back rate for incomplete reports	Upstream quality improvement, not just downstream speed
Trust	Override rate, within a two-sided band	Detects both an unhelpful model and a disengaged reviewer
Trust	Serious issues surfaced only by rules and missed by the model	The clearest early-warning signal of model drift
Trust	Campus-scoping violations detected in output	A single breach is a launch blocker, not a metric to optimise

Pilot targets

Metric	Target
Reports where the reviewer accepted the AI severity unchanged	≥ 70%
Median reviewer time per campus report	≥ 40% below the pre-pilot baseline
Time from a serious issue being written to being escalated	Same working day (baseline: one to two days)
Reports sent back for missing information	≥ 30% reduction

Go / no-go thresholds, fixed before the pilot

Condition	Threshold	If not met
P1 severity recall on the weekly labelled sample	≥ 0.90 sustained over four weeks	Do not expand. Improve rubric and grounding, re-run.
Ungrounded summary statements	≤ 2% of sampled statements	Tighten grounding constraints; suspend summaries above 5%.
Reviewer override rate	Between 10% and 25%	Below 10%: investigate complacency directly. Above 25%: the model is not adding value yet.
Campus-scoping or permission violations	Exactly zero	Immediate stop. Absolute launch blocker.
Inference cost per campus per month	Within the modelled ceiling	Reduce context size and batch, or restrict to serious items only.
Reviewer confidence (structured feedback)	Majority report the queue changed their working order for the better	Treat as workflow-fit failure, not model failure; redesign the surface.

Why the override band is two-sided

A very low override rate is not a success signal. It is the most likely early symptom of automation complacency, which is the specific failure this design is most afraid of.